

Srishyla Educational Trust ®

GM UNIVERSITY

(Established under the Karnataka State Act No. 19 of 2023) Post Box No. 4, Davanagere-577006



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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Academic Year 2025 - 26

Project Based Learning

Digital Notice Board

Semester: 2nd

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1.ABSTRACT

A Digital Notice Board is an electronic display system designed to replace traditional paper-based notice boards by providing a faster, more efficient, and environmentally friendly method of sharing information. The system uses digital screens such as LED, LCD, or smart displays to present notices, announcements, schedules, event information, and other important updates in real time.

The primary objective of a digital notice board is to enable instant communication between administrators and users. Information can be updated remotely through a computer, mobile application, or web interface, eliminating the need for manual posting and reducing maintenance efforts. The system improves accessibility, enhances visibility, and ensures that users receive the latest information without delays. The Digital Notice Board System can display a variety of content formats, including text messages, images, videos, scrolling announcements, event schedules, emergency alerts, and promotional materials. The integration of multimedia elements enhances user engagement and increases the visibility of important information. The system can also be configured to display notices according to predefined schedules, allowing administrators to automate the publication and removal of content based on specific dates and times. Abstract of Digital Notice Board System.

The Digital Notice Board System is an innovative and automated communication platform designed to display notices, announcements, events, and important information electronically. The main objective of this system is to replace traditional paper-based notice boards with a digital solution that enables quick, efficient, and real-time information sharing. In many educational institutions, offices, organizations, and public places, the manual process of updating notices is time-consuming and often results in outdated information being displayed. The Digital Notice Board System addresses these issues by providing a centralized platform for managing and displaying notices instantly.

This system allows authorized users or administrators to create, edit, update, and delete notices through a computer or web-based interface. The updated information is displayed immediately on digital screens, ensuring that users always have access to the latest announcements. The system can support various types of content, including text messages, images, videos, and scrolling announcements, making communication more attractive and effective.

2.INTRODUCTION

In the modern digital era, the rapid exchange of information has become an essential requirement for educational institutions, corporate organizations, government offices, hospitals, transportation facilities, and public service centers. Effective communication plays a vital role in ensuring that important announcements, notifications, schedules, and updates reach the intended audience accurately and on time. Traditionally, notice boards have been used as a primary medium for displaying information. While this method has been widely used for many years, it suffers from several limitations, including delayed updates, increased paper consumption, high maintenance effort, limited accessibility, and the possibility of outdated information remaining displayed for extended periods.

With the advancement of information and communication technologies, organizations are increasingly adopting digital solutions to improve communication efficiency and information management. One such innovation is the Digital Notice Board System, which provides a modern and automated approach to displaying and managing notices electronic. The system allows administrators to create, edit, schedule, and distribute notices remotely through a centralized platform, ensuring that users always have access to the latest information. Introduction to Digital Notice Board System

The Digital Notice Board System is a modern communication platform that uses digital technology to display notices, announcements, messages, and important information electronically. In today's fast-paced world, organizations require a quick and efficient method of sharing information with a large number of people. Traditional notice boards, which depend on printed papers and manual updates, often consume time, effort, and resources. To overcome these limitations, the Digital Notice Board System provides an automated and paperless solution for information dissemination.

The primary purpose of a Digital Notice Board System is to ensure that information reaches the intended audience quickly and effectively. It allows administrators or authorized users to create, update, and manage notices through a computer or web application. Once the information is entered into the system, it is displayed instantly on digital screens, making communication more efficient and reliable. This feature is especially useful in schools, colleges, universities, offices, hospitals, shopping malls, railway stations, airports, and other public places where regular updates are necessary.

3.PROBLEM STATEMENT

Traditional notice boards are widely used in schools, colleges, offices, hospitals, and public institutions for displaying announcements and important information. However, these notice boards rely on manual processes for creating, printing, posting, updating, and removing notices. This approach is time-consuming, labour-intensive, and often leads to delays in communication. Important information may not reach the intended audience promptly, especially when urgent updates or emergency announcements need to be communicated.

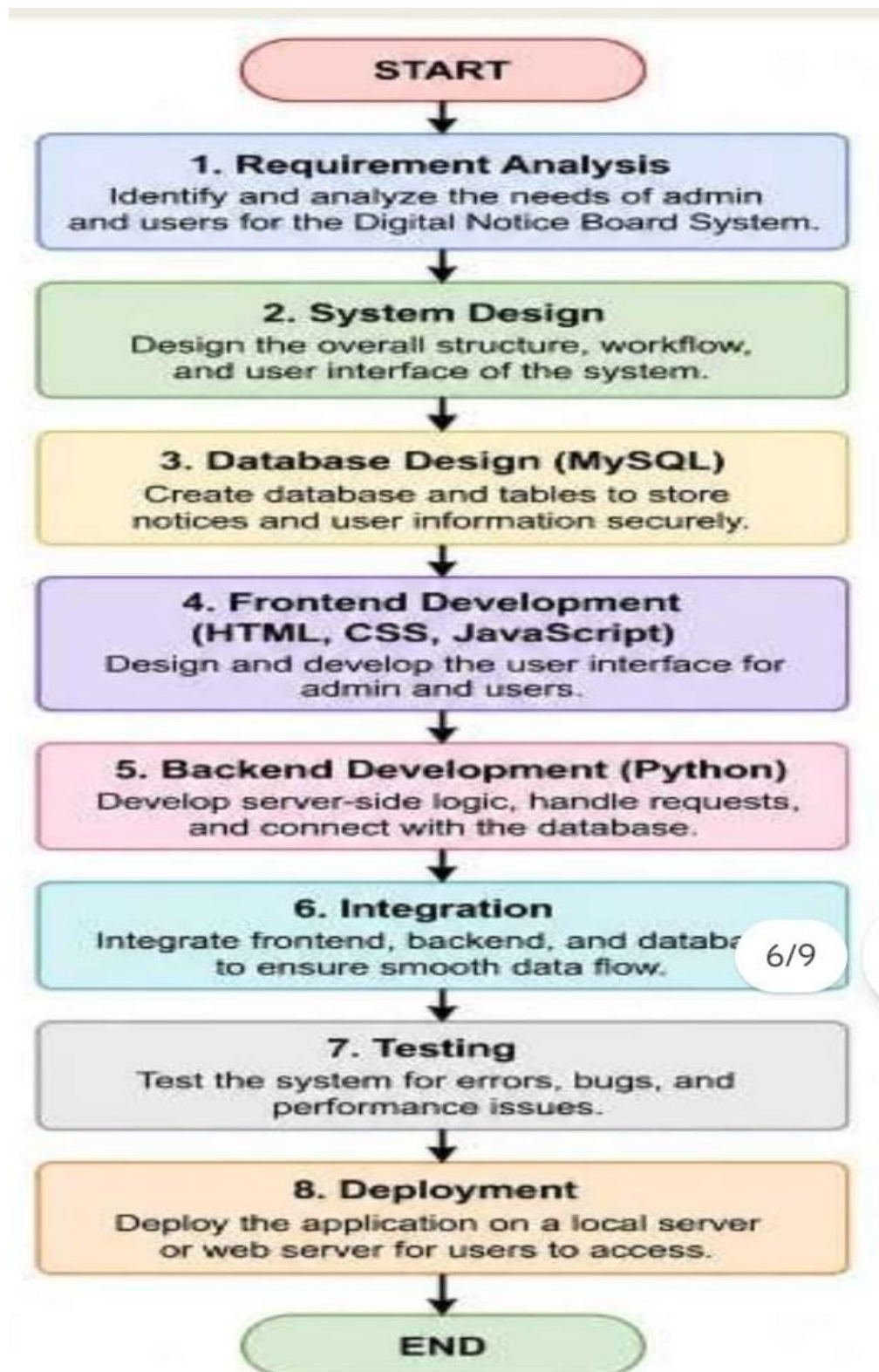
Additionally, traditional notice boards require continuous use of paper, printing materials, and physical maintenance, resulting in increased operational costs and environmental waste. Managing multiple notice boards across different locations can also be difficult, as each board must be updated separately. In many educational institutions, offices, organizations, and public places, important information is still communicated through traditional paper-based notice boards. Although these notice boards have been used for many years, they suffer from several limitations and challenges. The process of creating, printing, and displaying notices manually is time-consuming, costly, and inefficient. As the number of notices increases, managing and updating information becomes difficult, leading to communication delays and reduced effectiveness.

One of the major problems with traditional notice boards is the lack of real-time updates. Whenever new information needs to be displayed, an authorized person must physically visit the notice board location and replace the existing notice with a new one. This process can result in delays, especially when urgent announcements need to be communicated immediately. As a result, students, employees, or visitors may miss important information due to late updates.

4.OBJECTIVE

- **To automate the notice display process** by eliminating the need for manual printing and posting of notices.
- **To provide real-time updates** so that important announcements can be displayed instantly without delays.
- **To enable remote management of notices** through a web-based or mobile-based interface, allowing administrators to update content from any location.
- **To reduce paper consumption and printing costs**, thereby promoting environmentally friendly and sustainable communication practices.
- **To improve communication efficiency** by ensuring that users receive accurate and up-to-date information at the right time.
- **To support multimedia content**, including text, images, videos, and scrolling messages, making announcements more attractive and engaging.
- **To provide centralized control** for managing multiple display units from a single administration panel.
- **To enhance information accessibility** by displaying notices continuously in visible public locations.
- **To ensure security and authorized access** by allowing only approved users to create, edit, or delete notices.
- **To minimize maintenance efforts** associated with traditional notice boards and manual notice management.

5.METHODOLOGY



6.TOOLS & TECHNOLOGIES

1. Hardware: ESP32, LED Display

- **ESP32** is a microcontroller with built-in Wi-Fi that controls the digital notice board and receives updates wirelessly.
- **LED Display** is used to show messages, announcements, images, or scrolling text to users.
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2. Programming: C++ / Python

- **C++** is used to program the ESP32 and manage the display operations.
- **Python** can be used for developing the server-side application or controlling the notice board through a Raspberry Pi.
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3. Development Tools: Arduino IDE, VS Code

- **Arduino IDE** is used to write, compile, and upload code to the ESP32.
- **Visual Studio Code (VS Code)** provides an environment for developing web applications and backend programs with support for multiple programming languages.
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4. Communication: Wi-Fi

- Wi-Fi enables wireless transmission of notices from a computer or mobile device to the digital notice board, allowing real-time updates without physical connections.
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5. Frontend: HTML, CSS, JavaScript

- **HTML** defines the structure and content of the notice board interface.
- **CSS** is used to design and format the appearance of the displayed notices.
- **JavaScript** adds interactivity and enables dynamic updating of content.
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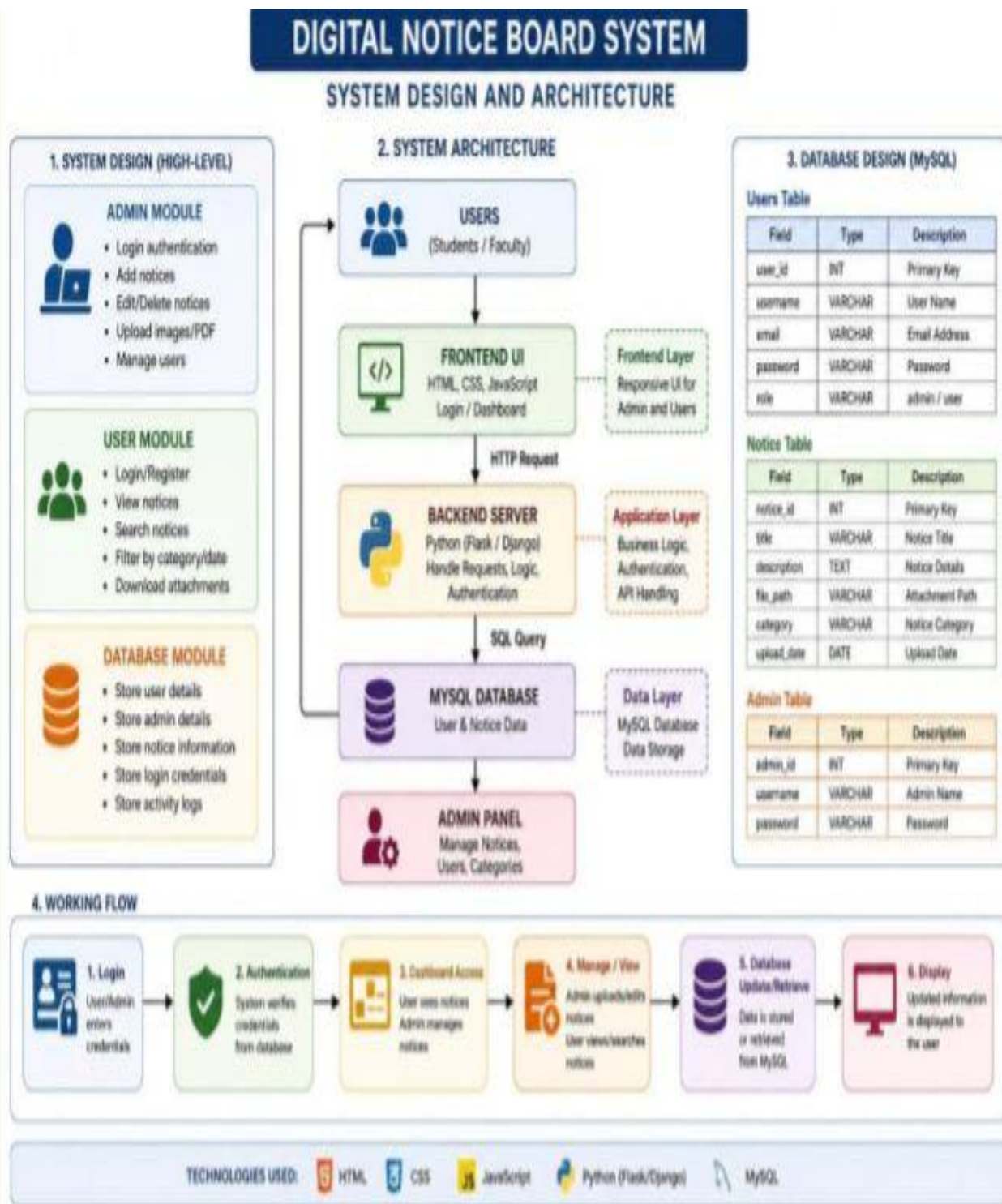
6. Database: Firebase (Optional)

- Firebase is a cloud-based database used to store notices and synchronize updates across devices in real time. It allows remote access and management of displayed content.
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7. Platform: Windows or Raspberry Pi OS

- **Windows** provides a development environment for writing and testing the software.
- **Raspberry Pi OS** runs on a Raspberry Pi and serves as the operating system for hosting and managing the digital notice board application.

7.SYSTEM DESIGN AND ARCHIETECTURE



8.IMPLEMENTATION

- Requirement analysis is conducted to identify system needs and user requirements.
- System architecture and workflow are designed.
- Administrator login module is developed for secure access.
- Database is created to store notices and announcements.
- User-friendly interface is designed for managing notices.
- Notice upload, edit, update, and delete features are implemented.
- Support for text, images, videos, and scrolling messages is added.
- LED/LCD display screens are connected to the system.
- Network connectivity is established for data transfer.
- Real-time notice updating functionality is implemented.
- Testing is performed to verify system functionality and security.
- Errors and bugs are identified and corrected.
- The system is deployed in schools, colleges, offices, or public places.
- Administrators are trained to operate the system effectively.
- Regular maintenance and software updates are carried out.
- Performance monitoring is conducted to ensure smooth operation.
- Security measures are implemented to protect data.
- Backup and recovery mechanisms are established.
- The system continuously displays updated information to users.

9.RESULTS AND OUTCOMES

Admin Dashboard

Add Notice

Semester Exam Timetable Released

Students are requested to check the latest semester exam schedule.

Exams

2026-06-10

Campus Placement Drive

Top companies are visiting for campus recruitment.

Placements

2026-06-15

Edit

Delete

College Cultural Fest

Annual cultural fest registrations are now open.

Events

2026-06-20

Edit

Delete

Digital Notice Board

4/6/2026, 12:42:16 pm

Professional Digital Notice Board

Stay Updated with Important Announcements

| Campus Placement Drive | College Cultural Fest

3

Total Notices

All Categories

General

Events

Exams

Placements

Urgent

All Categories

0

Urgent Notices

10. CHALLENGES AND LIMITATIONS

1. Internet Connectivity Issues

- Poor or unstable Wi-Fi can interrupt real-time updates.

2. Power Failures

- The system requires continuous power to function properly.

3. Limited Display Space

- Large amounts of information cannot be displayed simultaneously.

4. Hardware Limitations

- Devices like ESP32 have limited memory and processing power.

5. Security Risks

- Unauthorized access may lead to tampering with displayed content.

6. Maintenance Requirements

- Regular software updates and hardware maintenance are necessary.

7. Scalability Challenges

- Managing multiple displays and users becomes more complex as the system grows.

8. Data Synchronization Delays

- Notices may not update instantly due to network latency.

9. Environmental Factors

- Dust, heat, and moisture can affect the performance of outdoor displays.

10. Cost of Deployment

- Setting up and maintaining the system can be expensive for large installations.

- **LIMITATIONS**

- ❖ **Dependence on Internet Connectivity**

- Real-time updates require a stable Wi-Fi connection.

- ❖ **Limited Display Capacity**

- Only a limited amount of information can be displayed at one time.

- ❖ **Hardware Constraints**

- Microcontrollers such as ESP32 have limited memory and processing power.

- ❖ **Continuous Power Requirement**

- The system becomes unavailable during power outages.

- ❖ **Limited Multimedia Support**

- Basic systems may not support high-resolution images or videos efficiently.



- ❖ **Security Vulnerabilities**

- Without proper authentication, unauthorized users may access or modify notices.

- ❖ **Maintenance Overhead**

- Regular software updates and hardware maintenance are required for reliable operation.

- ❖ **Scalability Issues**

- Expanding the system to multiple locations may increase complexity and cost.

- ❖ **Environmental Sensitivity**

- Dust, moisture, and extreme temperatures can affect hardware performance.

- ❖ **Initial Setup Cost**

- Purchasing displays, controllers, and networking equipment can be expensive for large-scale implementations.

11.CONCLUSION

The Digital Notice Board System is an effective and modern solution for sharing information quickly and efficiently. It overcomes the limitations of traditional paper-based notice boards by providing real-time updates, remote access, and easy management of notices. The system helps organizations communicate important announcements, schedules, and alerts in a faster and more organized manner. By reducing paper usage and printing costs, the Digital Notice Board System also promotes an environmentally friendly approach to communication. Its ability to display multimedia content such as text, images, and videos makes information more attractive and easier to understand. The system improves accessibility, saves time, and enhances communication between administrators and users.

The Digital Notice Board System is a reliable, cost-effective, and user-friendly technology that meets the communication needs of modern educational institutions, offices, and public organizations. With future advancements in digital technology and networking, its applications and effectiveness are expected to grow even further, making it an essential tool for efficient information dissemination. The Digital Notice Board System is a modern and efficient solution for displaying announcements, notices, events, and important information electronically. It replaces traditional paper-based notice boards, making communication faster, easier, and more reliable. The system allows administrators to update notices instantly, ensuring that users always receive the latest information without delays. One of the major advantages of a digital notice board is its ability to save time, paper, and maintenance costs. Information can be updated remotely from a computer or mobile device, eliminating the need to print and manually replace notices. This makes the system environmentally friendly and cost-effective.

The Digital Notice Board System also improves the visibility and accessibility of information. Notices can be displayed in an attractive format using text, images, videos, and animations, which helps capture the attention of viewers. It is especially useful in educational institutions, offices, hospitals, railway stations, shopping malls, and other public places where timely communication is important. In addition, the system provides better organization and management of notices. Important announcements can be scheduled, prioritized, and displayed automatically. This reduces human effort and minimizes the chances of missing or outdated information being shown. The Digital Notice Board System is a smart, efficient, and user-friendly communication platform that enhances information sharing. By providing real-time updates, reducing paperwork, improving communication efficiency, and supporting modern digital technologies, it serves as an effective tool for organizations and institutions in today's digital world. It contributes to better communication, increased productivity, and a more organized environment.